Gwyneth Margaux Tangog (Gwyn)

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EDUCATION

Massachusetts Institute of Technology **Cambridge, MA**

Bachelor of Science in Computer Science and Engineering & Mathematics GPA: 4.9/5.0; Expected Graduation: May 2026

Masters of Engineering in Computer Science Expected Graduation: May 2027

Relevant Coursework: Software Construction, Design and Analysis of Algorithms, Machine Learning, Advances in Computer Vision

INDUSTRY EXPERIENCE

Recidiviz **June 2024 - August 2024**

Implementation Engineer Intern Remote

- Created raw data views and ingested raw data into the system for usage downstream.
- Implemented system schema changes to improve the organization and usability of data downstream.

The Engine Room **January 2024**

Data and Tech Intern Remote

- Performed data analysis on recent reporting data from the Light Touch Support program. Wrote automation scripts for repetitive data analysis tasks. Made and deployed an interactive data analysis dashboard.
- Github: https://github.com/gwyntangog/LITS_data_analytics Website: https://lits-analyst.onrender.com/

RESEARCH EXPERIENCE

MIT Theory of Computation Group **January 2025 - Current**

Undergraduate Researcher Cambridge, MA

- Developed hardness reduction proofs for the hardness of graph threading on grid graphs.
- Wrote polynomial-time algorithms and reductions to prove previously-unknown counting hardness of the solo chess problem.
- Wrote the manuscript *Counting Complexity of Unlimited Solo Chess*, which contributes new reduction techniques.
- Applied Exponential Time Hypothesis (ETH) to find lower bounds for the polyomino tiling problem.

MIT Concrete Sustainability Hub **September 2023 - June 2024**

Undergraduate Researcher Cambridge, MA

- Analyzed data using Python and prepared and presented a research poster to industry stakeholders on *The Effect of Gamification on Concrete Delivery Professionals*, exemplifying increases (25%-55%) in employee retention, productivity, and punctuality.
- Wrote Python scripts and data visualization scripts, and prepared and presented a research poster to industry stakeholders for the *Carbon Uptake* project, which models the carbon uptake potential of crushed concrete (fivefold of current).
- Modelled data and wrote related literature review in the published paper *Analysis of the impact of automaker strategies on lithium price elasticity using a novel bottom-up demand model* (https://doi.org/10.1016/j.resconrec.2025.108477).

MIT Economics Department **June 2023 - September 2023**

Undergraduate Researcher Cambridge, MA

- Cleaned and visualized data using STATA for the project *Heterogeneity of Child Penalty* to quantitatively relate child penalty to job flexibility.

TEACHING EXPERIENCE

MIT Electrical Engineering and Computer Science Department **September 2024 - Current**

Undergraduate Teaching Assistant Cambridge, MA

- Led a recitation section of 20 students and held office hours for the class “Math for Computer Science” (6.1200).
- Initiated and led review sessions for the class before exams, and other resources, such as an essay on graph induction.

MIT Interphase Edge Program **June 2025 - August 2025**

Residential Facilitator Cambridge, MA

- Led recitation sections both in-person and online, held office hours, and graded assignments and exams for Calculus II.
- Initialized the standardization of Calculus recitation material for future TAs to use on Overleaf.
- Mentored students and organized cluster outings as a cluster leader to promote a sense of community.

Awards: Outstanding UROP (Undergraduate Research Opportunity) Award 2024, MIT Eta Kappa Nu (HKN) Honors Society